PROJECT REPORT

Machine Learning Based Modeling of Antenna Using CST Studio

Submitted by

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Abstract

In this project, a custom machine learning model is developed to predict the S11 parameter of an antenna. The dataset is generated using CST simulations, which include antenna length, gap, and frequency. Instead of using standard machine learning algorithms, a custom mathematical model is designed and trained using gradient descent. The model successfully learns the behavior of the antenna and predicts S11 values with good accuracy. The results show that the machine learning model closely matches CST simulation results, reducing the need for repeated simulations.

Keywords: *Machine Learning, Antenna, S11, CST, Gradient Descent, Prediction Model*

1. Introduction

Antenna design is an important part of communication systems. Simulation tools like CST are widely used but can take time when multiple designs are tested.

A half-wave dipole antenna is one of the most basic and widely used antennas in communication systems. It consists of a straight conductor with a total length equal to half of the wavelength (λ/2).

The antenna operates by radiating electromagnetic waves when an alternating current flows through it. At resonance, the length of the antenna is approximately equal to λ/2, which gives maximum radiation efficiency.

The S11 parameter represents the reflection coefficient of the antenna. A lower S11 value (more negative in dB) indicates better impedance matching and efficient radiation.

In this project, the length and gap of the half-wave dipole antenna are varied to study their effect on the S11 parameter. The generated CST data is then used to train a machine learning model.

Machine learning provides a faster alternative. Once trained, a model can predict antenna performance instantly without running full simulations. In this project, a custom machine learning model is used to predict the S11 parameter using antenna dimensions and frequency.

2. Project Progress

The project was carried out in multiple stages, starting from basic antenna simulation to final machine learning implementation.

2.1 Initial Phase using Altair FEKO

Initially, the antenna design and simulation were performed using Altair FEKO. A patch antenna was considered, and full-wave electromagnetic simulations were used to understand antenna behavior.

The goal was to generate simulation data and apply machine learning techniques to predict antenna performance parameters such as S11 and resonant frequency.

Although the simulation was successfully completed on FEKO, sir suggested us to move to CST Studio and develop our own ML code instead of using the in-built ML algorithms in FEKO.

2.2 Transition to CST Studio and Dipole Antenna

Based on the project requirements and guidance, the approach was modified. The design was shifted to CST Studio Suite, and a half-wave dipole antenna was used instead of a patch antenna. In CST:

- Dipole antenna was designed using defined Length and Gap
- Time-domain solver was used for simulation
- Parametric sweep was performed to generate large dataset

This allowed easier data generation and better control over antenna parameters.

2.3 Machine Learning Development

Initially, a Random Forest regression model was implemented to predict S11 using the CST dataset.

Later, as suggested, a custom machine learning model was developed. This model uses a mathematical equation with adjustable parameters and is trained using gradient descent.

This final approach provided better understanding of the relationship between antenna parameters and S11, while also maintaining good prediction accuracy.

2.4 Final Outcome

Finally, the project evolved from:

- Altair FEKO + Patch Antenna

to:

- CST Studio Suite + Half-Wave Dipole Antenna
- Custom Machine Learning Model developed from scratch

This transition improved both implementation simplicity and learning depth, leading to a successful prediction model for antenna S11 parameters.

3. Methodology

The dataset used in this project is generated using CST by sweeping different antenna parameters:

- Length is varied from 100 mm to 150 mm with 5 mm steps
- Gap values used are 10 mm, 15 mm, and 20 mm
- This results in 33 different antenna configurations
- The frequency was swept from 0 to 2 GHz with a step size of 0.006 GHz

For each configuration, a frequency sweep is performed to obtain S11 values. In total, around 10,000 data points are generated and used for training the machine learning model.

The model learns the relationship:

Length + Gap + Frequency → S11

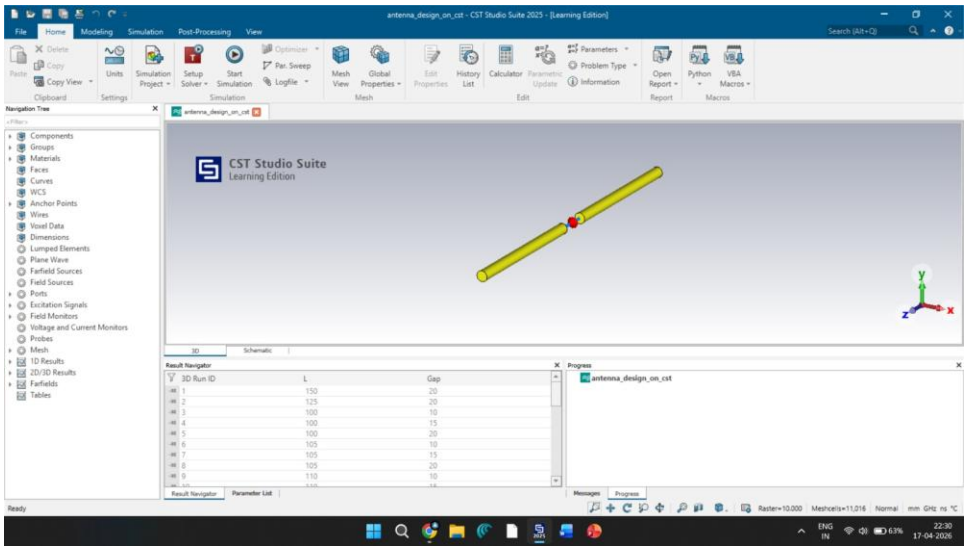


Figure 1: Half-wave dipole antenna designed in CST Studio Suite showing geometry and feed gap.

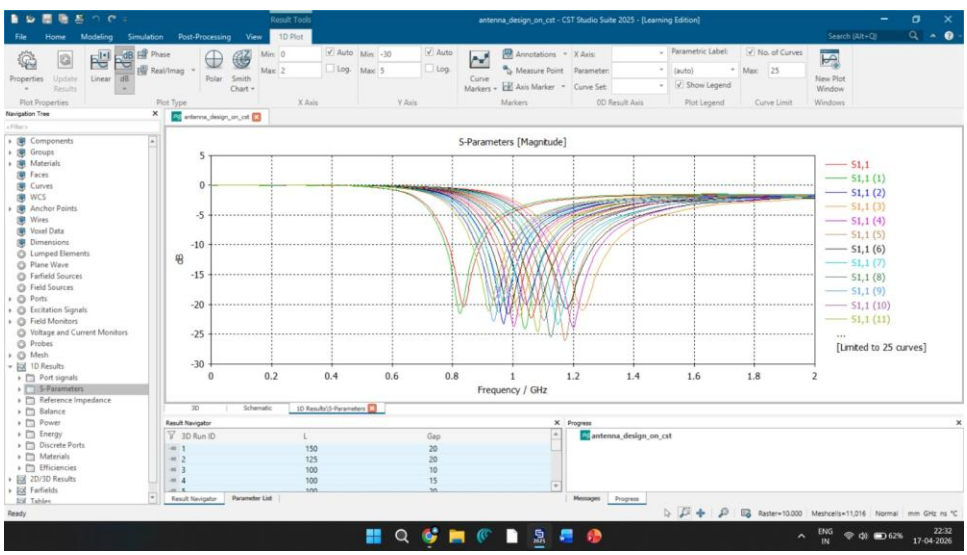


Figure 2: S11 parameter curves obtained from CST parametric sweep for different length and gap values.

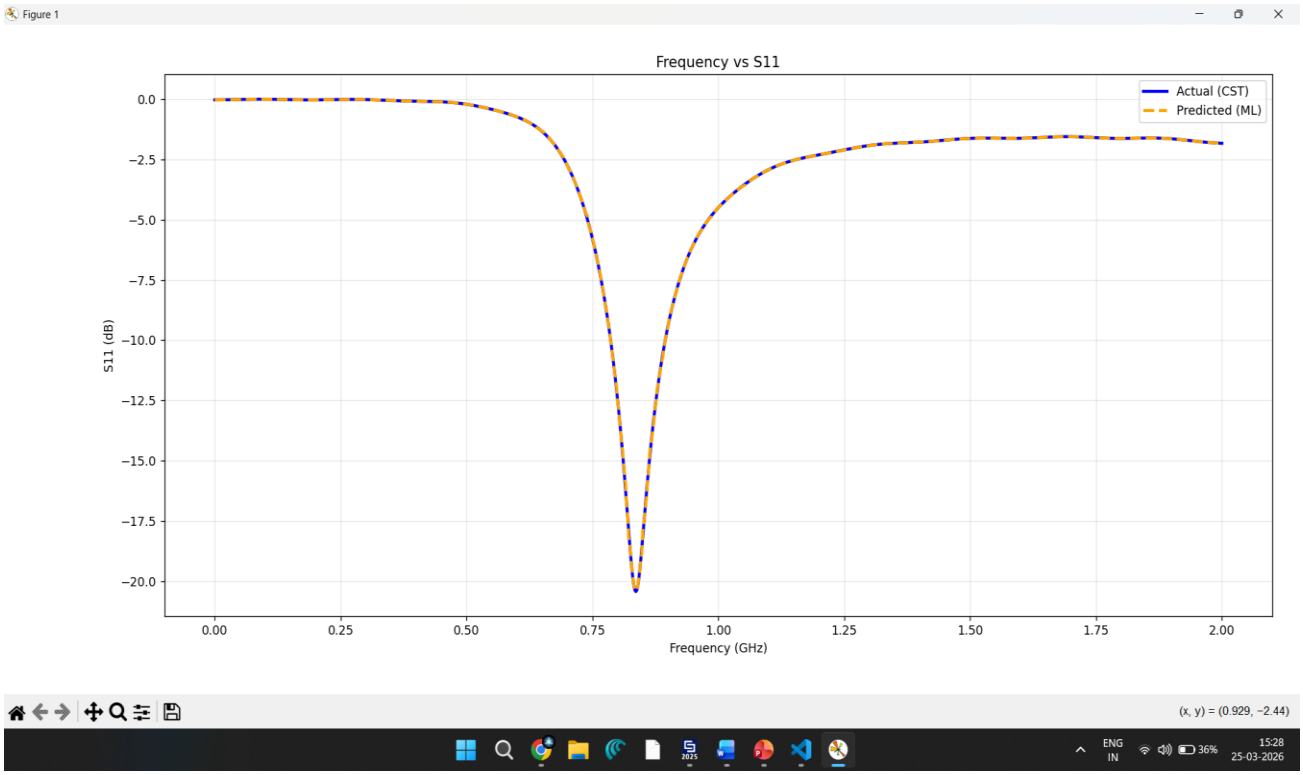


Figure 3: Comparison of actual CST S11 curve and Random Forest predicted S11 over frequency.

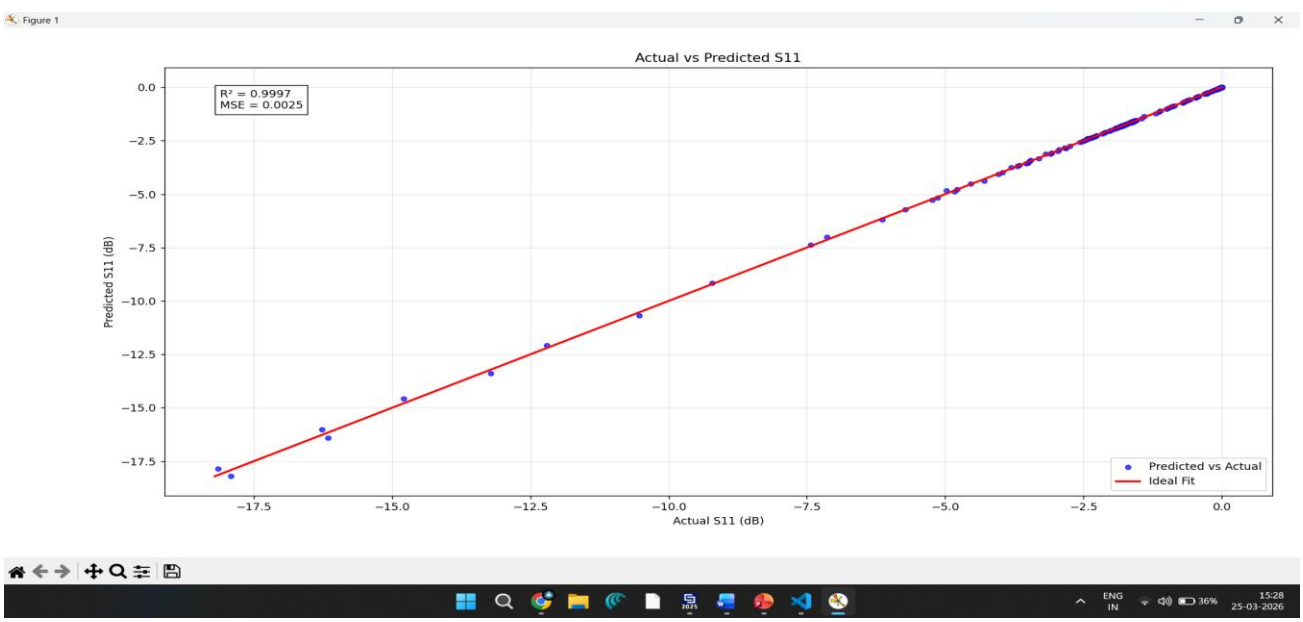


Figure 4: Actual vs predicted S11 values using Random Forest model.

RESULT and INTERPRETATION for this methodology:

1. **Frequency VS S 11 PLOT :**

- The predicted S11 curve closely follows the CST simulation across the full frequency range.
- The model accurately captures the resonance dip near ~0.8 GHz, including its location and shape.
- Minor deviation is observed near the minimum S11 value, indicating slight approximation error.
- Overall trend and nonlinear variation of S11 with frequency are well learned.

2. **Actual vs Predicted S11:**

- Data points lie very close to the 45° ideal line, showing strong agreement with CST results.
- $R^2 \approx 0.9997$ indicates the model explains almost all variation in S11.
- MSE ≈ 0.0025 confirms very low prediction error.
- Slight deviation is visible only at extreme S11 values near resonance.

4. Implementation

4.1 Dataset

The dataset is generated from CST simulations and contains approximately 10,000 data points. Each row includes:

- Length
- Gap
- Frequency (GHz)
- S11 (dB)

4.2 Data Preprocessing

All input values are normalized before training. This helps in faster training and improves stability.

4.3 Custom Machine Learning Model

Instead of using standard ML models, a custom mathematical model is created. The model includes:

- Sinusoidal terms to represent wave behavior
- Resonance terms to capture dips in S11
- Geometry terms for Length and Gap effects
- Interaction terms between inputs

4.4 Training Method

Training is done using gradient descent. The model updates parameters step-by-step to reduce error. The loss function used is Mean Squared Error (MSE):

$$MSE = (1/n) \times \Sigma(y_actual - y_predicted)^2$$

5. Results and Discussion

5.1 Model Performance

The trained model achieved the following performance metrics:

- MSE = 0.8833
- $R^2 = 0.9312$

This shows that the model has low error and good prediction capability.

5.2 Prediction Accuracy

The predicted values closely follow the actual CST values. Most points lie near the ideal diagonal line, showing good accuracy.

5.3 Frequency Response Analysis

For Length = 150 mm and Gap = 15 mm:

- Resonance dip around 0.8 GHz is captured correctly
- Overall curve matches CST results
- Slight deviation near minimum S11
- ML output is smoother than CST curve

Since the model is trained on a large dataset of 10,000 points, it generalizes well across different antenna configurations.

5.4 Effect of Varying Learning Rate and Epochs

To further evaluate model performance, experiments were conducted by varying the learning rate (lr) and number of training epochs. This helps analyze the impact of training dynamics on prediction accuracy.

Case 1: lr = 0.0020, epochs = 11000

1. Frequency vs S11 plot
2. Actual vs Predicted plot
3. Console/result image showing R^2 and MSE

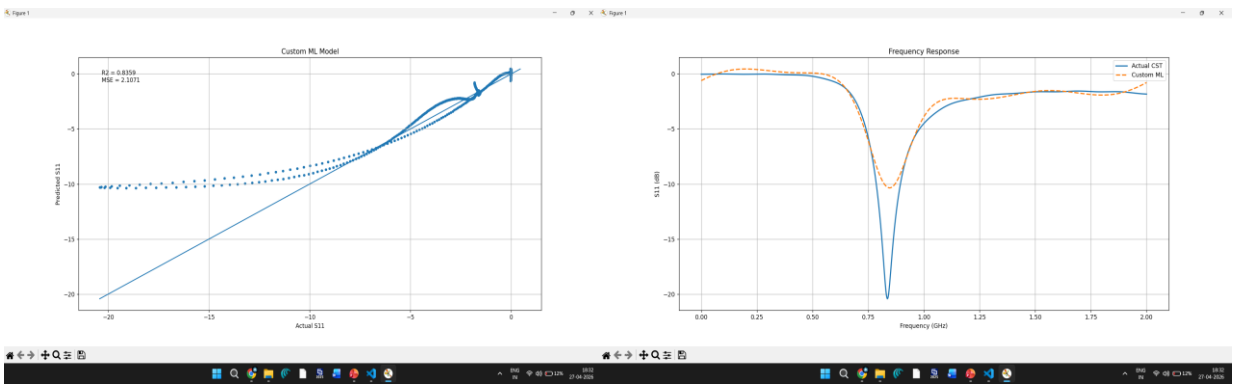


Figure 1: Results for lr = 0.0020 and epochs = 11000.

- The predicted curve follows the general CST trend but shows deviation near resonance.
- Resonance dip is captured, but depth is not accurately matched.
- **$R^2 \approx 0.839$, MSE ≈ 2.107**, indicating moderate prediction accuracy.

Inference:

- Lower epochs lead to incomplete learning of nonlinear S11 behaviour.
- Model shows signs of underfitting.

Case 2: lr = 0.0023, epochs = 19000

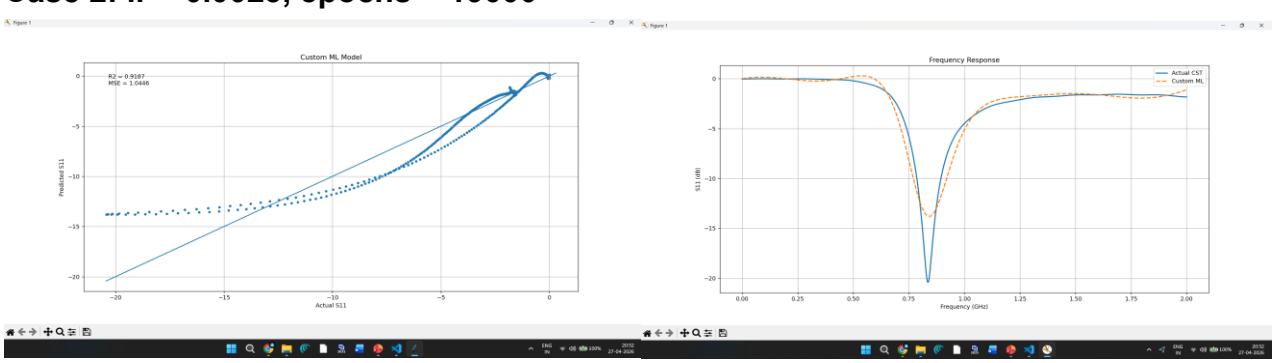


Figure 2: Results for lr = 0.0023 and epochs = 19000.

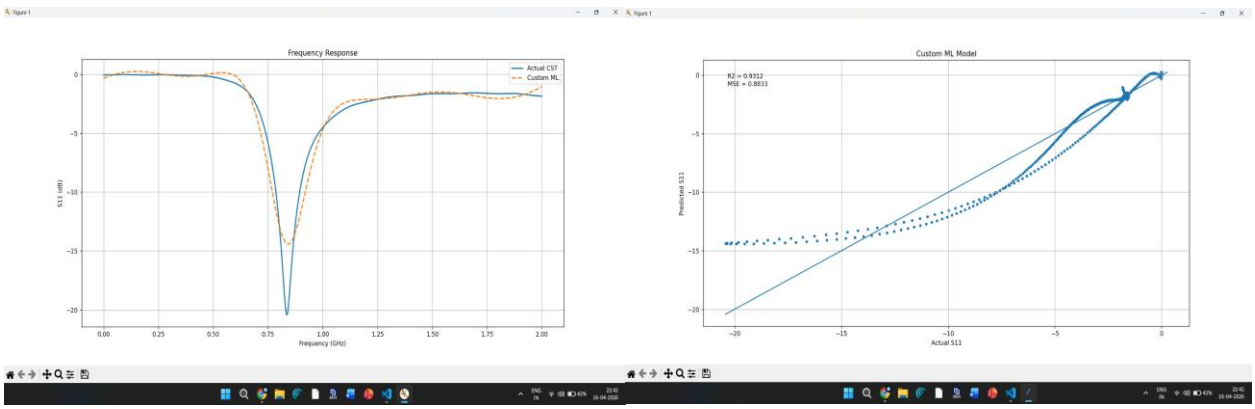
- Improved alignment with CST curve compared to previous case.
- Resonance location is captured more accurately, with slight amplitude mismatch.

- Minor fluctuations observed due to higher learning rate.
- **$R^2 \approx 0.839$, MSE ≈ 2.107** (or your exact values if slightly different).

Inference:

- Increasing epochs improves learning of nonlinear patterns.
- Higher learning rate introduces small instability in predictions.

Case 3: lr = 0.0015, epochs = 20000



s*Figure 3: Results for lr = 0.0015 and epochs = 20000.*

- Predicted curve closely matches CST simulation across full frequency range.
- Resonance dip (~0.8 GHz) is accurately captured in both location and depth.
- Smooth curve indicates stable convergence.
- Shows lowest error among all configurations.

Inference:

- Lower learning rate enables fine tuning of model parameters.
- Higher epochs ensure complete learning of S11 nonlinear characteristics.
- Provides best accuracy and generalization.

6. Conclusion

In this project, a machine learning based approach was successfully developed to predict the S11 parameter of a half-wave dipole antenna. The antenna was designed and simulated using CST Studio Suite, and a large dataset of around 10,000 samples was generated using parametric sweep of antenna length, gap, and frequency.

Initially, a Random Forest model was used for prediction, but later a custom machine learning model was developed as suggested. This model was able to learn the relationship between antenna parameters and S11 effectively.

The results show that the predicted S11 values closely match the CST simulation results, with an R^2 value of 0.9312 and low mean squared error. The model also successfully captures important features such as the resonance dip and overall frequency response of the antenna.

Overall, the project demonstrates that machine learning can be effectively used to model antenna behavior and reduce the need for repeated electromagnetic simulations. This approach helps in faster analysis and can be useful for future antenna design optimization.

7. Future Scope

- Use deep learning models for better accuracy.
- Increase the datasets for improved performance and analysis.
- Extend to other antenna parameters like gain and radiation pattern.

References

[1] CST Studio Suite, Dassault Systèmes, "CST Studio Suite Documentation," 2025.

[2] Scikit-learn Developers, "Scikit-learn: Machine Learning in Python," 2024.

[3] C. A. Balanis, Antenna Theory: Analysis and Design, 4th ed., Wiley, 2016.

[4] D. M. Pozar, Microwave Engineering, 4th ed., Wiley, 2012.

[5] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.